DEMO / SYNTHETIC: Energy density, keyhole porosity and hardness of selective laser melted AlSi10Mg

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Abstract

DEMO / SYNTHETIC TEXT generated to test the AM-KG pipeline; the numbers are not real research results. AlSi10Mg samples were produced by selective laser melting (SLM) using a laser power of 0.35 kW and scan speeds between 1000 and 1600 mm/s. A volumetric energy density above 70 J/mm^3 caused keyhole porosity. T6 heat treatment reduced the hardness of the alloy.

1. Introduction

Aluminium alloys processed by SLM show fine cellular microstructures.

2. Experimental procedure

AlSi10Mg powder was processed on an SLM machine with a layer thickness of 30 µm and a hatch spacing of 130 µm. The volumetric energy density ranged from 40 to 90 J/mm3. The build plate temperature was 150 °C. Relative density was measured with the Archimedes method, and the microstructure was characterized by optical microscopy and SEM. Hardness was measured using a Vickers hardness tester.

3. Results

The highest relative density of 99.4 % was obtained at an energy density of 60 J/mm^3. Increasing the scan speed reduced the volumetric energy density and increased lack of fusion. Energy densities above 70 J/mm^3 promoted keyhole porosity. The as-built hardness was 125 HV0.1. The as-built microstructure showed a cellular Si network along melt pool boundaries.

4. Discussion

The T6 heat treatment dissolved the cellular Si network and decreased the hardness to 95 HV. The laser power did not affect the surface roughness in the investigated range.

5. Conclusion

Volumetric energy density influenced keyhole porosity and lack of fusion in SLM AlSi10Mg.